

LANCASTER, PA, USA

WMO: 725116

Lat: 40.120N

Lon: 76.294W

Elev: 123

StdP: 99.86

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54737

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.6	-10.3	-20.2	0.6	-11.4	-17.7	0.8	-8.8	12.6	-0.4	11.6	-1.9	3.7	290	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	32.8	23.9	31.4	23.1	30.0	22.4	25.2	30.7	24.4	29.1	23.6	28.0	4.0	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	18.8	27.6	22.9	17.9	26.6	22.3	17.3	26.1	77.6	30.4	73.9	29.0	71.0	28.1	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.1	8.0	-15.9	35.0	2.4	1.9	-17.6	36.3	-19.0	37.4	-20.3	38.5	-22.0	39.8		
(5)			-16.6	26.5	2.2	1.0	-18.2	27.2	-19.5	27.8	-20.7	28.4	-22.3	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	-1.1	0.2	5.0	11.1	16.8	21.7	24.0	23.0	19.4	12.8	6.9	1.8	(6)
(7)		DBStd	9.79	5.55	5.19	5.09	4.65	4.23	3.30	2.73	2.75	3.67	4.54	4.50	4.79	(7)
(8)		HDD10.0	1224	345	277	169	39	2	0	0	0	0	22	114	257	(8)
(9)		HDD18.3	2902	602	508	413	221	82	9	1	2	31	180	343	511	(9)
(10)		CDD10.0	1908	1	2	14	73	213	352	433	404	281	109	21	3	(10)
(11)		CDD18.3	544	0	0	0	5	34	111	174	148	62	9	0	0	(11)
(12)		CDH23.3	4650	0	1	4	72	329	950	1585	1186	458	64	1	0	(12)
(13)		CDH26.7	1509	0	0	0	15	91	299	586	381	126	11	0	0	(13)
(14)	Wind	WSAvg	3.2	3.7	3.8	4.0	3.8	3.1	2.9	2.5	2.3	2.5	3.0	3.3	3.5	(14)
(15)	Precipitation	PrecAvg	1114	80	65	96	92	90	105	107	103	108	97	78	91	(15)
(16)		PrecMax	1575	165	145	164	198	150	282	213	198	337	226	144	159	(16)
(17)		PrecMin	752	29	13	26	31	26	30	30	38	10	9	23	16	(17)
(18)		PrecStd	180	35	30	36	40	34	51	45	42	77	57	33	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	18.7	23.2	28.8	31.6	33.3	35.6	34.0	32.4	28.0	21.2	17.8	(19)
(20)			MCWB	13.7	13.8	15.4	18.9	21.9	23.6	24.7	24.3	23.3	21.2	15.9	14.9	(20)
(21)		2%	DB	12.8	13.9	18.9	25.7	29.1	31.6	33.2	32.2	29.9	25.0	18.7	14.2	(21)
(22)			MCWB	10.2	9.8	13.0	17.0	20.9	22.9	24.4	23.8	21.9	19.7	14.8	12.0	(22)
(23)		5%	DB	9.1	10.2	16.2	22.4	27.2	29.9	31.5	30.4	27.8	22.6	16.6	11.4	(23)
(24)			MCWB	7.1	6.4	11.1	15.5	19.8	22.2	23.5	22.9	20.7	18.2	13.3	9.6	(24)
(25)		10%	DB	6.7	7.4	13.2	19.6	24.8	28.1	29.9	28.8	26.1	20.7	14.1	8.8	(25)
(26)			MCWB	4.5	4.6	9.0	13.8	18.7	21.1	22.7	22.2	19.9	16.5	11.2	6.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	15.0	16.6	20.6	23.5	24.9	27.1	26.1	25.0	22.9	17.8	15.8	(27)
(28)			MCD B	17.8	17.6	21.6	25.2	29.4	31.1	32.9	31.8	30.1	25.1	19.9	17.1	(28)
(29)		2%	WB	10.5	10.4	14.2	18.5	22.0	24.0	25.5	25.1	23.5	21.0	15.9	12.6	(29)
(30)			MCD B	12.0	13.3	17.5	23.5	27.4	29.3	31.1	30.2	26.9	24.1	17.8	14.1	(30)
(31)		5%	WB	7.5	7.0	12.0	16.6	20.8	23.2	24.5	24.1	22.5	19.1	13.9	9.9	(31)
(32)			MCD B	9.0	9.2	15.2	21.2	25.5	28.0	29.5	28.2	25.8	21.9	16.0	11.2	(32)
(33)		10%	WB	4.7	4.8	9.5	14.6	19.6	22.4	23.8	23.3	21.5	17.1	12.0	6.9	(33)
(34)			MCD B	6.4	7.4	13.0	18.8	23.7	26.8	28.3	27.0	24.7	19.5	13.8	8.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.0	8.8	10.1	11.3	11.0	10.4	10.3	10.2	10.5	10.2	9.5	8.0	(35)
(36)			MCD BR	11.4	12.9	14.6	15.3	13.8	12.2	12.2	11.8	12.6	12.6	12.2	10.8	(36)
(37)			MCWBR	9.6	9.6	9.7	8.8	6.8	5.2	4.8	4.8	5.7	7.0	8.5	9.3	(37)
(38)		5% WB	MCD BR	10.7	11.6	13.3	13.5	12.2	10.8	10.3	9.9	9.8	10.5	9.9	10.2	(38)
(39)			MCWBR	9.6	9.6	9.9	8.6	6.6	5.3	4.7	4.6	5.2	6.7	8.1	9.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.326	0.358	0.392	0.431	0.497	0.499	0.499	0.415	0.365	0.326	0.307	(40)	
(41)		taud	2.435	2.363	2.356	2.286	2.211	2.066	2.072	2.059	2.264	2.414	2.487	2.484	(41)	
(42)		Ebn at Noon	855	887	893	881	851	793	786	773	822	833	829	830	(42)	
(43)		Edh at Noon	80	99	112	128	142	164	162	160	122	94	76	72	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.86	2.75	3.67	4.75	5.32	5.87	5.94	5.18	4.25	3.03	2.14	1.59	(44)	
(45)		RadStd	0.16	0.34	0.28	0.43	0.64	0.48	0.30	0.30	0.54	0.33	0.20	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+150	N/A
(47) Regional (72 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+123	+69

Nomenclature: See separate page